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An illustrative JVP trace is presented in Fig.3 where the waves a, c, x, v and y are visible. The parameters a_i , x_i , v_i and y_i for each cardiac cycle i were identified by the algorithm. However, about 10 % of the points are normally not properly detected by the algorithm, while it was possible to identify them manually (see Fig. 4). The parameters a_i , v_i , t_{a_i} and t_{v_i} were calculated as in Eq. (2). The mean values and standard deviations of the parameters are presented in Table 1. JVP and ECG time relationship The time interval between the wave a and the events P and R and the wave x and the event T are calculated for each cardiac cycle. These intervals are denoted as: $t_{aR_i} = t_{a_i} - t_{R_i}$ $t_{aP_i} = t_{a_i} - t_{P_i}$ $t_{xT_i} = t_{x_i} - t_{T_i}$ (3) Table 1: Example of mean value and standard deviation of main JVP parameters are presented. a x v a v t ax tvy HR (cm²) (cm²) (cm²) (cm²) (s) (s) Bpm 1.57 0.18 1.29 0.16 1.33 0.18 0.29 0.04 0.04 0.02 0.33 0.06 0.11 0.05 61 Table 2: Example of mean value and standard deviation of time intervals t_{aR} and t_{xT} averaged over several cardiac cycles. t_{aR} t_{xT} t_{aP} (s) (s) (s) 0,03 0,02 0,12 0,06 0,14 0,05 Figure 2: The figure shows an illustrative JVP trace during a cardiac cycle together with the ECG trace. The waves a, c, x, v and y of the JVP